Aviation taxation proposals

July 2026

1 Introduction

This appendix argues for:

1. a set of aviation tax reform proposals
2. a unilateral tax policy, using any component from 1), that earmarks at least a certain fraction (e.g., 60%) of revenues to global funds for global public goods or global redistribution, automatically dividing the tax rates by 2 for any flight connecting to another country already having in place such a policy.
3. A solidarity levy coalition mechanism that would build on 2) and formalize the norm that it supports: coalition members would respect the norm that unilateral extra-territorial taxation of flights is only legitimate if at least a certain fraction (e.g. 60%) of revenue gets earmarked to global funds whose eligibility for receiving funding would be determined via a representative vote (e.g. population-weighted qualified majority voting at the UN). Moreover, coalition members would implement the conditional policy defined by 2), thereby incentivizing other countries to join the coalition.

The EU could implement 2) and initiate 3). Simply implementing unilateral extraterritorial carbon pricing for flights with full revenue retention, as it did in 2012, would risk triggering opposition by other countries that will likely again consider it to be a breach of existing international norms. By instead implementing 2) and initiating 3), the EU could establish a legitimate architecture that would not only address the externalities associated with flights but also create a stable source of funding for truly additional global public good provision and global redistribution.

2 Tax reform proposals

The following table summarizes the four tax instruments that we propose should ideally all be used[1]:

tax instrument	roles	estimated optimal rates
tax on vacant seats	achieve socially optimal load factor	63% of average ticket prices
carbon tax	internalize climate damages	\$ 210/tCO2 (EPA) multiplied by 3
VAT	ensure the same treatment as other consumption expenditure	17.5 %
tax on busines class	counteract the lack of progressivity of overall tax system	?

Table 1: Caption

2.1 Tax reforms to transport the same number of people with fewer and fuller planes

We propose the following tax reform as a minimal reform that could garner broad international support:

[1]Not having the VAT as an instrument is likely to impose little costs, given that one could simply adjust the carbon tax by an amount equivalent to the average VAT rate applied to the rest of the economy.

Definition 1 *The passenger-flow-preserving efficiency-maximizing tax reform consists of introducing the following tax instruments: 1) a tax on vacant seats, whereby airlines have to pay for each vacant seat a certain percentage of the average ticket price on the route 2) a carbon tax*

The reform consists of setting the tax on vacant seats to achieve that the socially efficient vacancy rate gets reached and adjusting the carbon tax such that the number of passengers being transported remains unchanged as compared under the status quo.

Dörpinghaus et al. 2026 provide a full micro-economic model that aims to accurately capture the following tradeoff: taxing vacant flights allows for transporting the same number of people with fewer planes but it makes it more likely that flights sell and no more people can buy tickets even if they have a very high willingness to pay. Dörpinghaus at al. 2026 show that the market outcome tends to leave too many seats empty. The model disregards distributional questions to focus on efficiency. However, taking into account distributional effects would typically strengthen the case for taxing vacant seats. The calibrated model yields the following results for the passenger-flow-preserving efficiency-maximizing reform:

1. The tax on vacant seats is 63 percent of average sold ticket price.
2. The additional carbon tax is \$95/tCO2.
3. The vacancy rate gets reduced from 20% to 3%.
4. Flight frequency reduces by 17%.
5. Emissions (before accounting for fuel market leakage) decline by 0.17*0.8*900=122 MtCO2 if introduced globally and 0.29*122=35MtCO2 if introduced on all flights to and from EU countries.
6. Aviation tax revenue increases globally by \$35 billion if introduced globally and $0.29 \times \$35 billion = \$10 billion$ if introduced on all landing in and/or departing from EU countries.
7. Global economic surplus, without counting the avoided climate damages, increases by \$72 billion if introduced globally and $0.29 \times \$72 billion = \$21 billion$ if introduced on all flights to and from EU countries.
8. Assuming a SCC of \$210 and a fuel market leakage rate of 78% (`https://www.kielinstitut.de/fileadmin/Dateiverwaltung/IfW-Publications/fis-import/21d3bce2-f815-4875-8c01-cfd3e807213e-KWP_2283.pdf`), the avoided climate damages amount to $0.22 \times 210 \times 0.122$ billion = \$5.6 billion per year if introduced globally. Taking into account the radiative forcing from contrails multiplies these avoided damages by a factor of 3 according to (https://www.sciencedirect.com/science/article/pii/S1352231020305689), thus resulting in \$16.8 billion per year.
9. The world market price for oil falls. For the EU, this yields a gain through lower import expenses on oil of $\$4.5 billion$ ((`https://www.kielinstitut.de/fileadmin/Dateiverwaltung/IfW-Publications/fis-import/21d3bce2-f815-4875-8c01-cfd3e807213e-KWP_2283.pdf`)), assuming global introduction.

2.1.1 Incidence

In general, aviation tax reforms raise the difficult question of how the incidence is split between the travelers and the hospitality sector. Conceptually, it is clear that this is determined by the relative elasticities of the demand for flights and the supply of hospitality services (see IMF Staff Climate Note CLNEA/2024/003). However, estimating these elasticities is difficult and, as a result, there appear to be no published studies giving any confident estimates. In the next section, we will provide our best estimates based on Knispel et al. 2026. However, to discuss the incidence of the passenger-flow-preserving tax reforms, we do not need to take a stance on these questions.

In fact, an attractive feature of the passenger-flow-preserving tax reforms is that, at least in the simplest economic models, they create no burden for the hospitality sector, i.e. for those providing goods and services to travelers such as hotels, restaurants and tourism service providers. Moreover, the impact on travelers (via changes in prices, changes in availability of flights and changes in flight frequency) is well-captured by the model from Dörpinghaus et al. 2026 and thus already taken into account in the above estimates.

2.2 Additional carbon taxes

Let us consider a carbon tax imposed by the EU on all flights arriving in or departing from EU countries. Here are the results based on the model from Knispel et al. 2026:

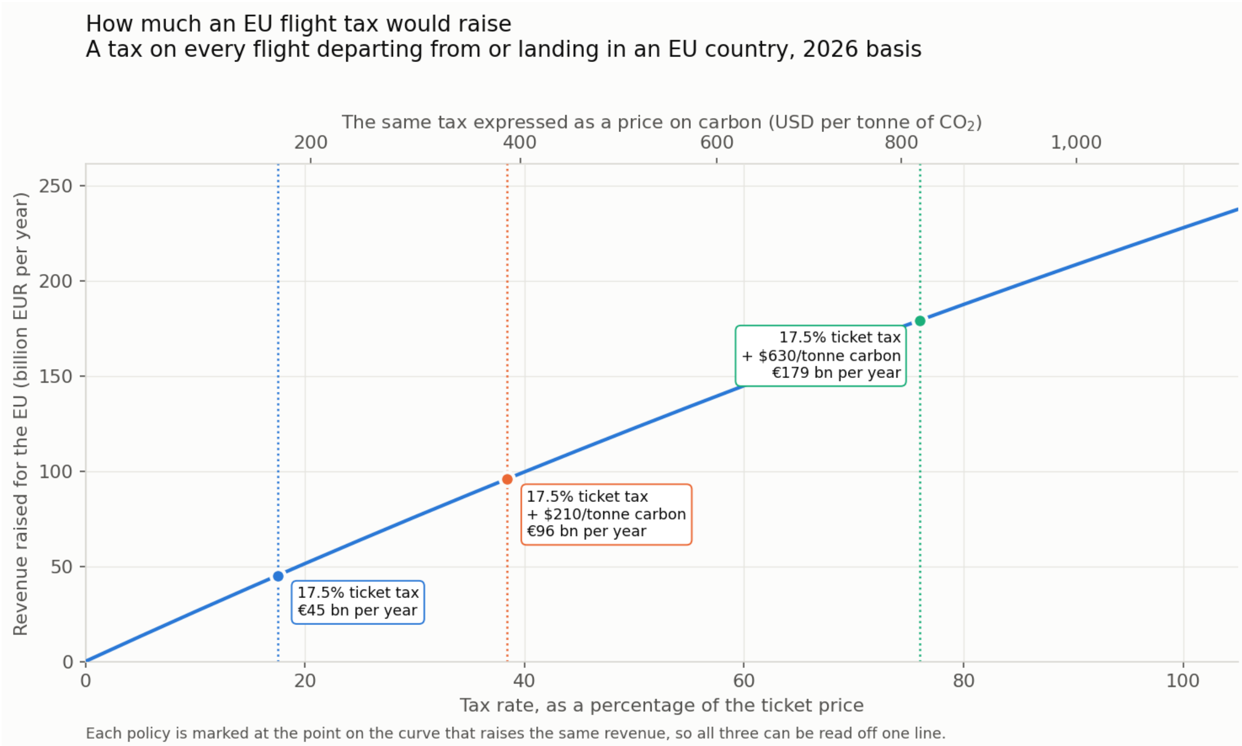

Figure 1: Enter Caption

The plot shows three levels for a carbon tax that could be put in place in addition to the above passenger-flow-preserving reform:

1) A carbon tax rate that compensates for the absence of VAT on international flights: classical Walrasian models imply that the overall optimal tax policy consist of setting a uniform VAT plus Pigouvian taxes[2]. To optimize tax policy in a given sector, such as aviation in our case, holding the other sectors' tax policies constant, one should set an ad valorem tax equal to the average ad valorem taxes (17.5% in the case of the EU) in the other sectors before adding the taxes that address externalities.

2) A carbon tax equal to the global social cost of carbon: Here we use the EPA's estimate of $\$210/tCO2$.

3) A carbon tax taking into account contrails: This multiplies the SCC by a factor of 3 according to (https://www.sciencedirect.com/science/article/pii/S1352231020305689).

The following plot shows the interaction between the VAT and the carbon tax: as we increase the carbon tax, people fly less, thus reducing the tax base and the VAT revenue. It might come as a surprise that this effect is not stronger. In fact, in the above plot, we see that even at an ad valorem tax rate of 100% (which, to be clear, is above the socially optimal rate according to the estimates of the SCC provided here) tax revenue is still increasing in the tax rate. This is based on a calibration with a price elasticity of demand for flights of -0.6 based on (cite IATA study or IMF).

[2] More precisely, the VAT plus the markup should be the same across sectors. Given that Markups have historically been 0 in the aviation sector and on average positive in the rest of the economy this means that the VAT should actually be higher.

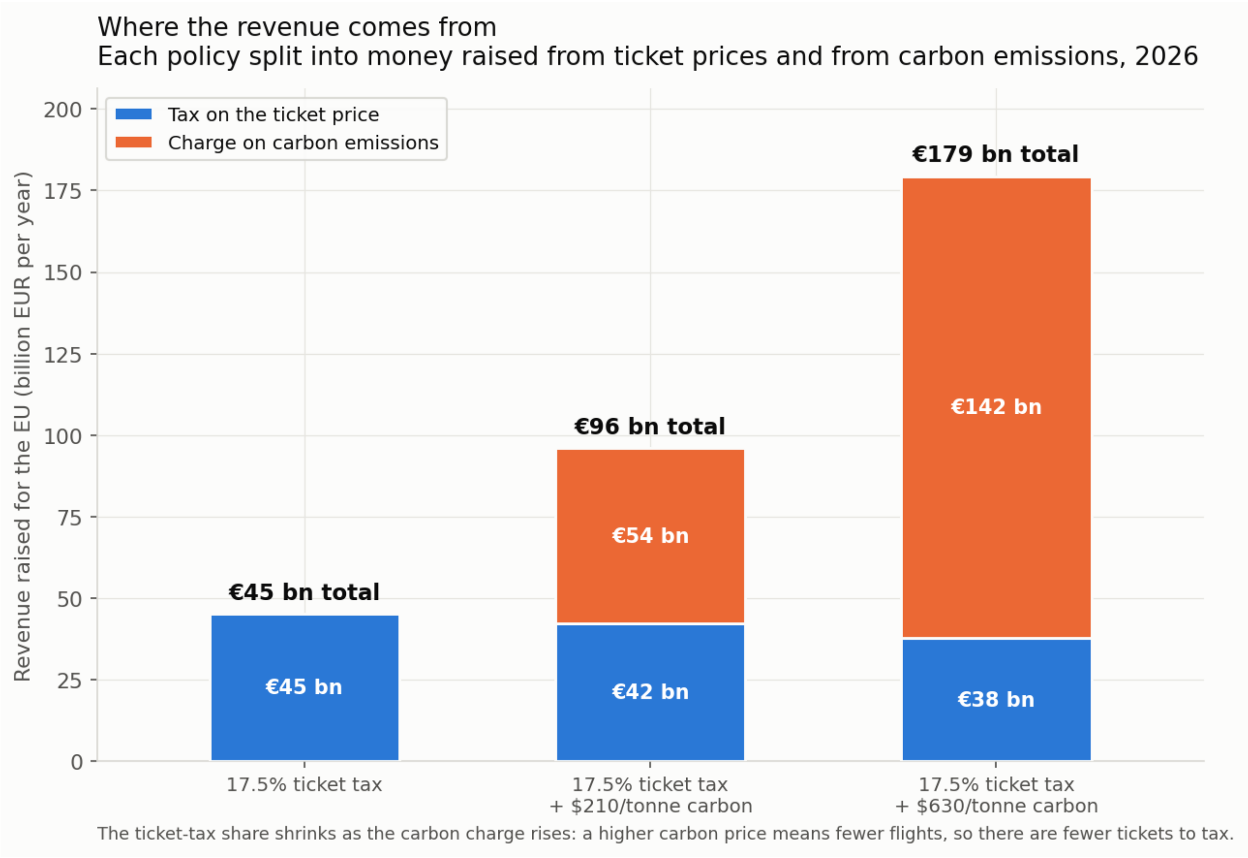

Figure 2: Enter Caption

2.2.1 Proportions of the carbon tax revenue collected by EU member states

The estimates of current emissions from Zheng et al. (...), provide us with an estimate of the shares of the tax revenues collected by each EU member state under the tax policies proposed above:

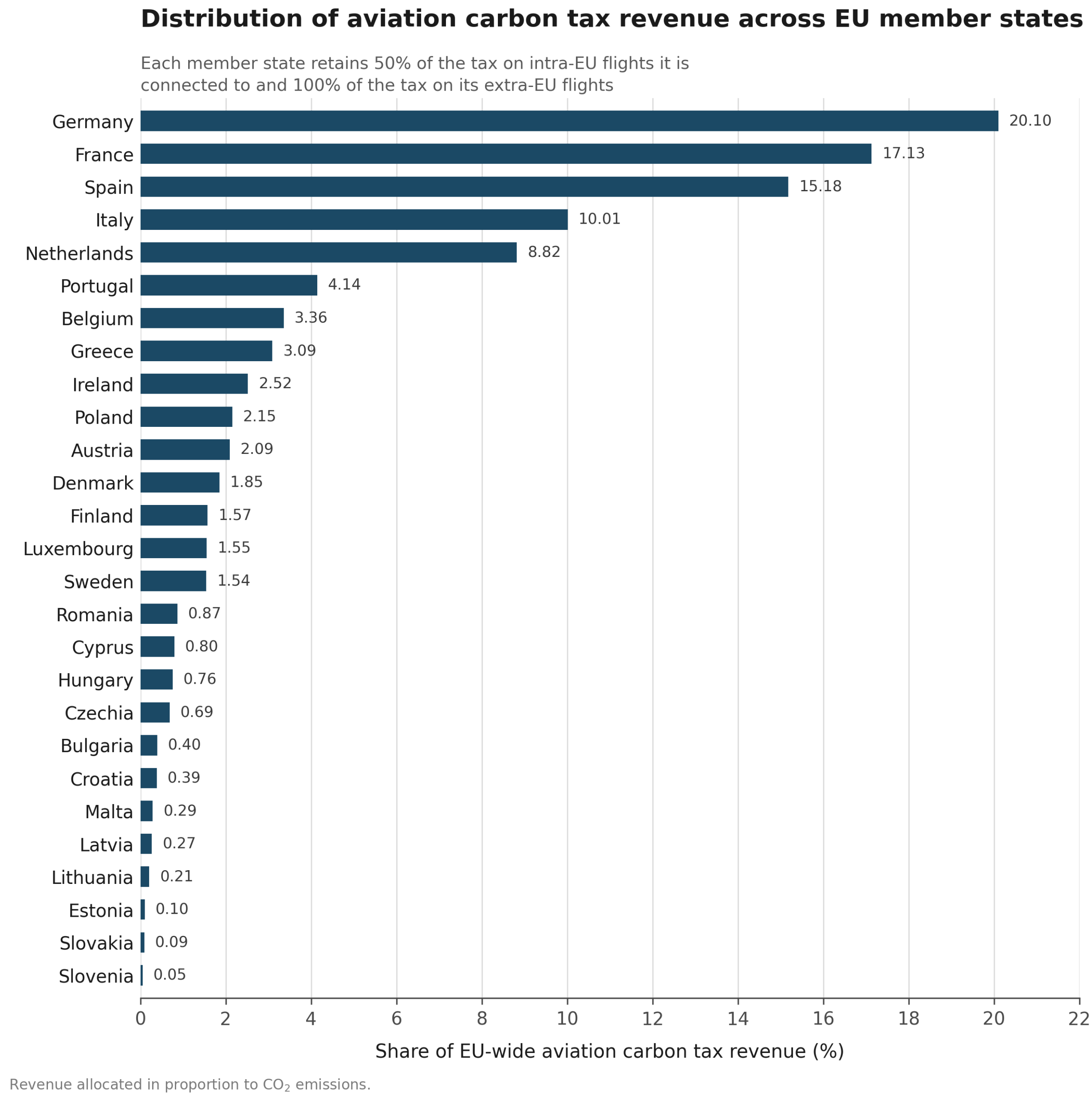

Figure 3: Enter Caption

These number are directly relevant for assessing the challenges that would arise under a proposal to forward there entirety of the ETS1 revenue corresponding to aviation to the EU budget. One possible assumption[3] is that this will simply displace the GDP-proportional required contributions that now cover any shortfall in funding for the EU budget. The following plot shows the ratio of carbon tax revenue to GDP, thereby revealing the winners (the ones with a ratio below 1) and losers (the ones with a ratio above 1) of such a reform:

[3]In reality, it is more plausible that the impact will be a mixture of such a displacement of GDP-based contributions and an expansion of the EU budget.

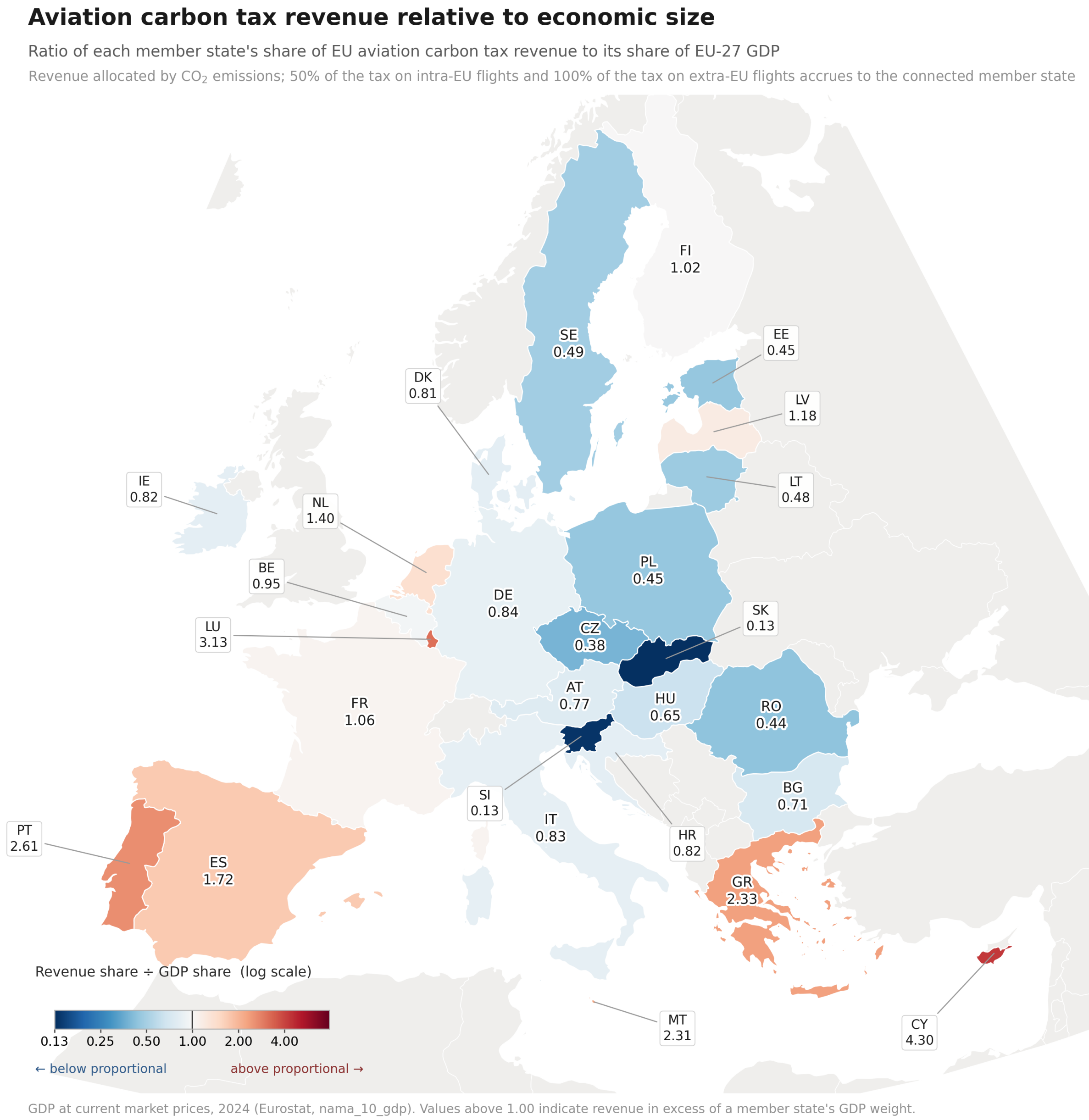

Figure 4: Enter Caption

Reforming the EU ETS requires a qualified majority among the member states in the Council of the European Union and a simple majority in the European Parliament under the standard ordinary legislative procedure. [1, 2]Qualified Majority Voting (QMV)In the Council of the European Union, decisions are passed using the "double majority" rule. To pass a reform, the following conditions must be met: [1]

- Member State Support: At least 55% of the member states must vote in favor (which means 15 out of the 27 countries).
- Population Representation: The voting countries must represent at least 65% of the total EU population. [1, 2]

Based on the above estimate and assuming self-interested voting behaviour, we find: The 55% state criterion is comfortably met ($17 \geq 15$), but the population criterion is not: 63.0% against a 65% threshold, short by 9.0 million people.

The results highlight the likely difficulties for a reform that would introduce a carbon tax on aviation (running in parallel to the EU ETS) as an EU own resource if the revenue simply replaces GDP-based contributions with *existing* ETS1 revenue from aviation: By itself, such a reform would likely have difficulties garnering unanimity. However, efficiency-enhancing tax reforms like those involving the introduction of a tax on vacant seats as discussed above would lower the actual tax burden and therefore make unanimity more likely. Moreover, extending taxation to include flights connecting the EU with other countries would be less costly for the EU due to the shifting of part of the tax burden to other countries, as we will demonstrate in the next section. However, we will also argue there that simply extending taxation /ETS1 extra-territorially in this way will likely trigger opposition and trade sanctions by other countries. This will motivate our proposal, detailed further below, for the EU to initiate an aviation solidarity levy coalition that would raise funding for global funds for global public goods and could incentivize most or even all other countries to join. We will later argue that this would be in the long term interest for the EU as it could reap large benefits from the resulting improved global public goods provision.

2.3 Global incidence

Here we present the results from the micro-economically founded global model from Knispel et al. 2026 with endogenous hospitality prices that allows us to take into account that the economic burden of aviation taxes falls partly on travelers and partly on the countries to whom they travel (specifically the hospitality sector in these countries). All the plots show the welfare effects without taking into account without taking into account the avoided climate damages which we discuss further below.

Recall that we are considering carbon taxes imposed by the EU on all flights arriving in or departing from EU countries.

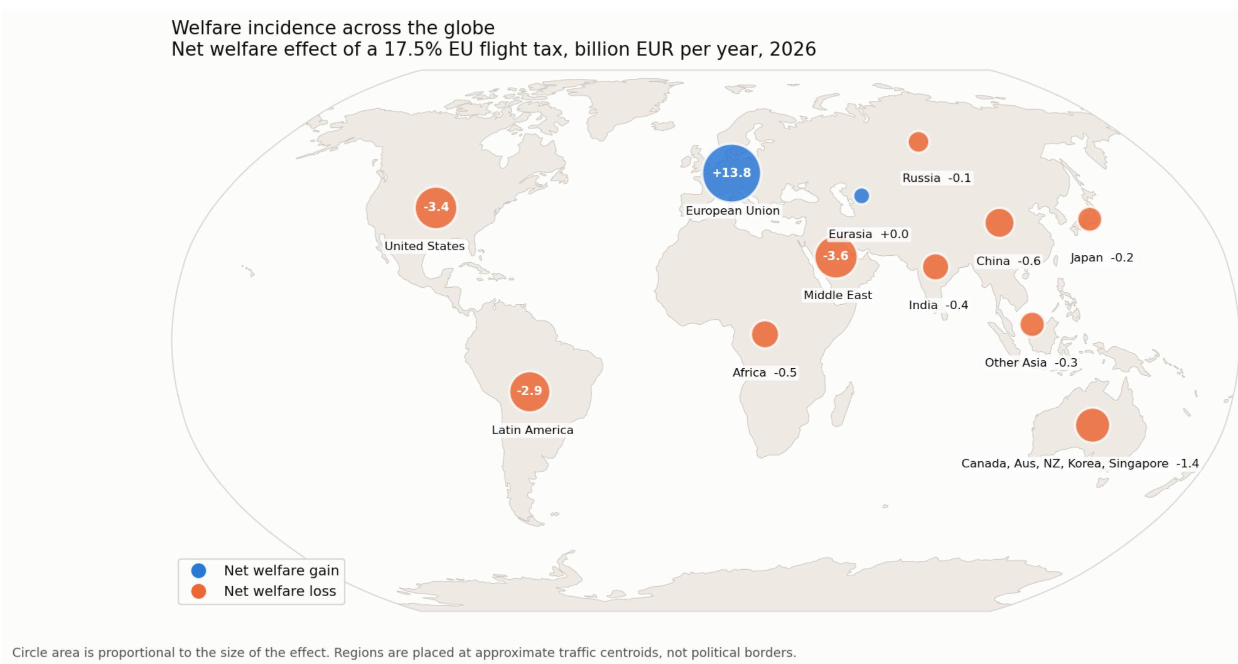

Figure 5: Enter Caption

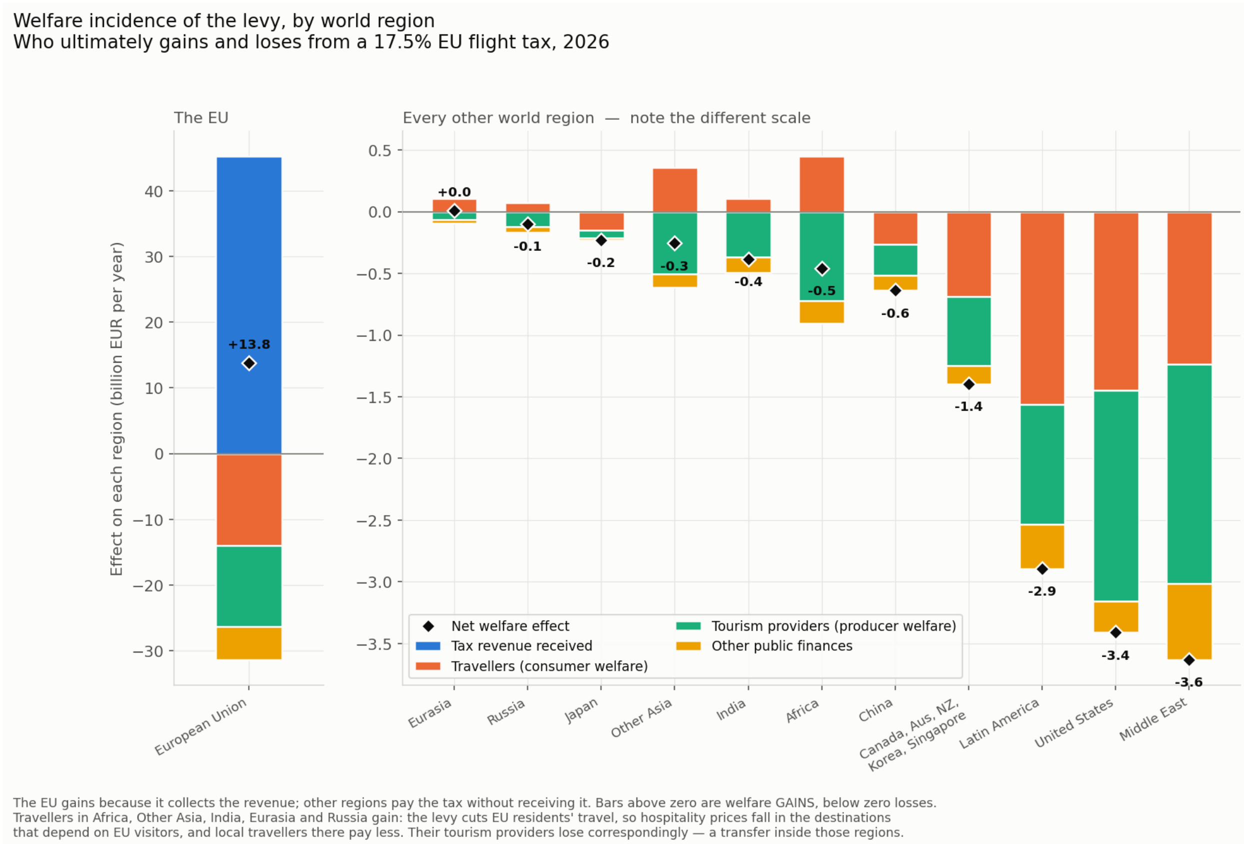

Figure 6: Enter Caption

Overall, we see that all other regions lose out on net. This can help explain why the EU faced severe opposition when it extended the ETS to all international flights in 2012 and retained all the revenue for itself.

Moreover, we see that the EU has unilateral incentives to unilaterally tax aviation. It turns out that the EU's welfare maximizing unilateral tax (assuming full revenue retention) is 99% percent in ad valorem terms. Similarly, other countries and regions have strong unilateral incentives for high aviation taxes:

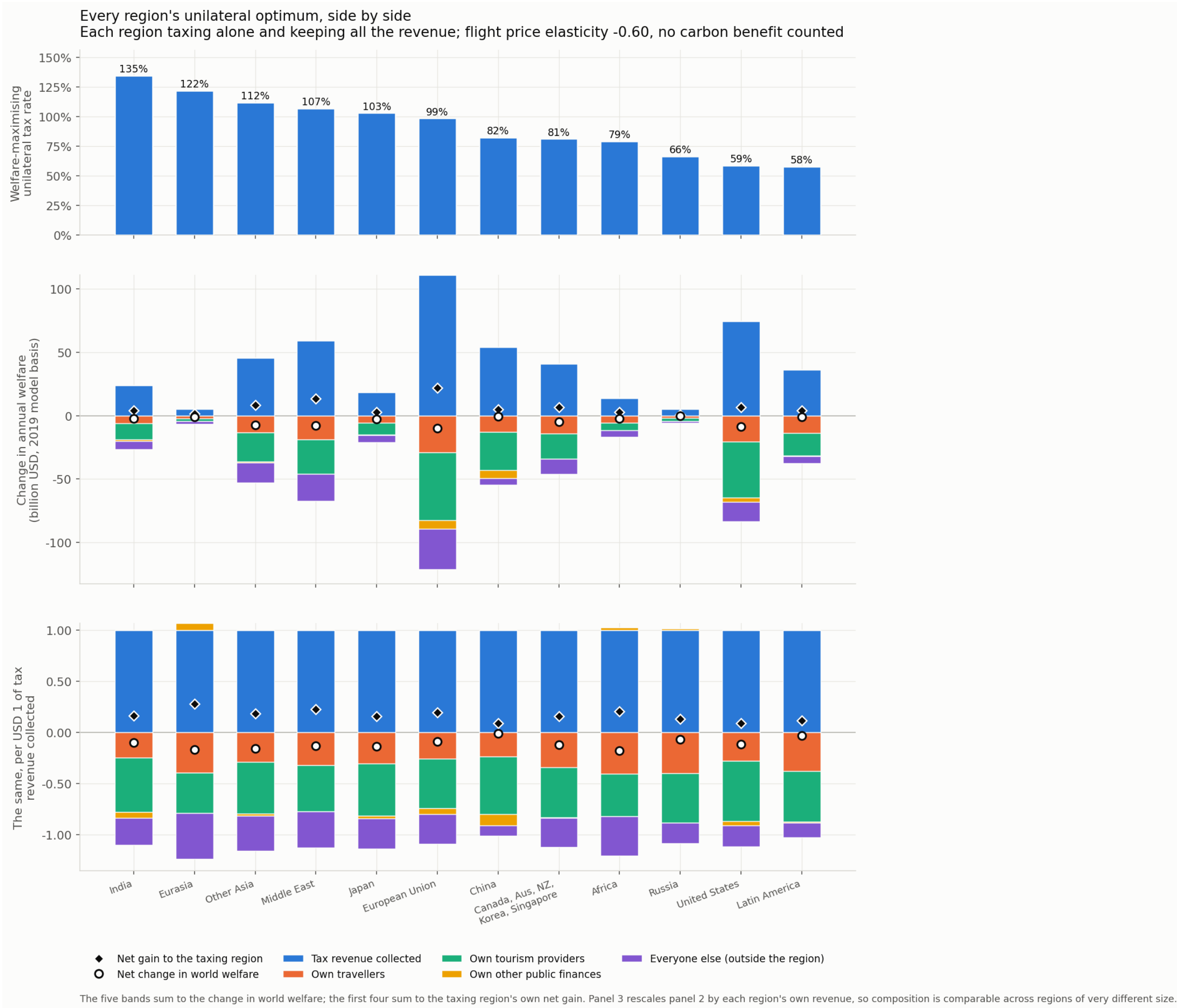

Figure 7: Enter Caption

We see that the unilaterally optimal tax rates exceed on average substantially the 78% that we saw above to internalize the full climate damages and correct for the currently missing VAT. Thus in the absence of the legal norms established by the Chicago convention and subsequent legal rulings against unilateral extraterritorial aviation taxation, the world might end up with excessive aviation taxes which could lead to substantial welfare losses, in particular if other hard-to-quantify external effects of increased travel based on the "contact hypothesis" and potentially decreased international conflict turn out to be large. [4].

At the very least, the above results show that the norm against unilateral extraterritorial taxation of flights had a strong rationale in 1944 when it was substantiated via the Chicago Convention, given that at the time the social cost of carbon was small and not even widely recognized. There should be a strong

[4] The intergroup contact hypothesis has a large and reasonably robust evidential base: the meta-analysis of Pettigrew and Tropp (2006) finds a mean contact–prejudice correlation of about -0.21, rising to roughly -0.33 in the more rigorous experimental designs. However, Allport's facilitating conditions — equal status, common goals, cooperation, institutional support — are systematically violated by mass leisure tourism. The tourism literature finds that service-oriented contact (tourist as customer, local as service provider) can *increase* perceived cultural distance even where social-oriented contact reduces it, and studies of structured all-inclusive package travel report increases in prejudicial attitudes towards hosts, consistent with the "cultural bubble" interpretation. Thus for tourism it is not so clear whether it is on net favorable for good inter-population relations (and thus, more speculatively, on inter-state relations). However, it is plausible that student mobility, professional and scientific collaboration, long-duration immersive stays and travel to visit family is beneficial for inter-population relations, and thus likely also for inter-state relations. An assessment of the size of these effects is beyond the scope of this report

presumption in favor of respecting existing international norms, particularly if it has a well-grounded global welfare rationale. As circumstances change (in this case, as the social cost of carbon has become large and well-recognized), existing norms should ideally be refined, not disregarded.

Rather than oppose the norm against unilateral extraterritorial taxation of aviation emissions, the EU should try to make it evolve, recognizing both the rationale for it and its problems in the current world. The norm against unilateral extraterritorial taxation of flights should have an exception for taxes that are earmarked to global funds legitimized via a UN vote, as we will propose below. One motivation for such a modified norm is that it could allow the world to generate a stable flow of funding for global funds for global public goods and global redistribution. The above incidence results provide a separate further motivation: if countries can at most retain a certain fraction (e.g. 40%) of their aviation tax revenue then they have less incentives to choose inefficiently high aviation taxes. Thus the risk of overtaxation that the incidence results highlight will still be contained via the modified norm.

Once such an exception has been added to the norm against unilateral extraterritorial taxation of flights, the norm should be strengthened to be applied to all forms of flights taxation, including ticket taxes that are currently excluded from it. In contrast to the norm against unilateral extra-territorial taxation itself, the exception for ticket taxes lacks any discernible rationale. In fact, the above results on the welfare gains through fuller planes achievable via taxes on vacant seats imply that ticket taxes have a negative effect on economic surplus, given that they incentivize higher vacancy rates.

Under such a modified norm that would apply to all forms of taxes and carbon prices, countries' incentives could align to move towards an efficient taxation of aviation, using a mixture of taxes on vacant seats to achieve socially optimal vacancy rates [5] and carbon taxes to internalize the climate externalities and to compensate for the lack of VAT. The overall level of taxation will then be somewhere between the current under-taxation of flights and the potential over-taxation of flights that the Chicago Convention was meant to guard against.

2.4 Progressive taxes on tickets

ICCT (2025), "Aviation levy" report (ID-302) provides different design options.

3 Conditional aviation tax policies with earmarking for global funds for global public goods

We propose to implement the following at the EU level:

Definition 2 *Let the "earmarking obligation" denote the rule to either earmark aviation tax revenue to global funds currently recognized as eligible (via a population-weighted vote at the UN) or retain the proportion r, where the proportion $1 - r$ then has to be allocated in proportion to how much the different eligible global funds receive.*

The conditional aviation solidarity levy at rate τ , denoted P^*, is defined by implementing all the conditional policies (i.e. P^* is "P_0 and $P_1$and P_2 and P_3 and P_4") defined as follows [6]:

P_0: Tax all domestic flights at rate τ and respect the earmarking obligation.

P_1: Tax all international flights at rate $\frac{\tau}{2}$ and respect the earmarking obligation.

P_2: For all international flights coming from or going to a country not doing P_0 or not doing P_1 increase the tax rate to τ and respect the earmarking obligation [7].

P_3: For all international flights coming from or going to a country not doing P_2 increase the tax rate to τ and respect the earmarking obligation.

P_4: For all international flights coming from or going to a country not doing P_3 increase the tax rate to τ and respect the earmarking obligation.

[5] and taxes on business class to partly offset the current lack of progressivity in income and wealth taxation

[6] Ideally, even further such conditional policies would be implemented, i.e. P_5 , P_6 and so on. We have truncated at P_4 for better readability.

[7] For ease of exposition, we consider here only the binary case. In practice, it would be better to use a more general version of the conditional policies, where if a country does P_0 and P_1 at a lower rate $\tau' < \tau$ then P_2 would increase the tax rate to $\tau - \frac{\tau'}{2}$. This is analogous to how the EU CBAM credits lower carbon prices abroad.

This conditional policy P^* creates an incentive for other countries to also adopt such a conditional policy. To see why, let us denote by the letter A the action "adopt the carbon price on aviation and earmark the revenue to global funds currently recognized as eligible (via a population-weighted vote at the UN) or retain the proportion r, where the proportion $1-r$ then has to be allocated in proportion to how much the different eligible global funds receive". The conditional policy P^* has the property of incentivizing other countries to adopt A, to incentivise yet others to adopt A and to incentivise others to incentivise others to adopt A. If the EU were to adopt P^* then any country also adopting P^* would be guaranteed to only get its flights connecting to the EU taxed at the halved rate $\frac{\tau}{2}$.

4 Aviation solidarity levy coalition mechanisms

Building on the conditional aviation tax policies with earmarking for global funds for global public goods proposed in the previous section, the EU could initiate the following international mechanism:

Definition 3 *The aviation solidarity levy coalition mechanism with minimal carbon price τ_0 is defined by the following rules:*

***Taxation obligation**: Tax all domestic flight emissions at some carbon price $\tau \geq \tau_0$ and all international flights at τ for all the connections with countries not in the coalition.*

For flights connecting with countries in the coalition, only apply the tax rate $\frac{\tau}{2}$.

***Earmarking obligation**: either earmark aviation tax revenue to global funds currently recognized as eligible (via a population-weighted vote at the UN) or retain the proportion r, where the proportion $1-r$ then has to be allocated in proportion to how much the different eligible global funds receive.*

***Voting right**: right to nominate any global fund for a vote on eligibility*

By initiating such a mechanism, the EU could likely avoid the opposition that it encountered in 2012 in response to the extension of the ETS1 to include all international flights. Back then, none of the carbon pricing revenue got allocated to other countries or to global funds. The incidence analysis shown above suggests that the rest of the world would have born much of the economic burden of the carbon price on the EU's international flights.

However, if the EU had allocated the money to global funds for climate change mitigation and other global public goods, the resulting benefits would likely have more than offset the burden of the carbon price for most of the other countries. This in itself would have made it less likely that other countries would have opposed the measure. In addition, a rules-based mechanism like the aviation solidarity levy coalition defined above could confer legitimacy to participants and further reduce the risk of opposition.

5 Implementation options

We have above proposed three tax instruments that could be implemented at the EU level if unanimity can be reached among member states. For the case of the carbon tax, it could alternatively be implemented via the ETS1 which has the advantage of not requiring unanimity.

If unanimity could be reached then it would be advantageous to implement additional EU-wide taxes on flights as proposed above. A tax on seats might most easily garner unanimous support if EU member states end up believing the above results on welfare gains through fuller planes. Moreover, an additive EU-wide carbon would be useful as a complementary instrument that can co-exist with the ETS1.

The conditional aviation tax policy with earmarking for global funds is most easily implemented for taxes. However, it can be adapted to the case of an ETS. Similarly the aviation levy solidarity coalition mechanism can be formulated more generally to allow for participation with an ETS or also with a mixture of an ETS and tax instruments like a tax on vacant seats. The taxation obligation would be defined based on the ratio of tax&ETS revenue collected over the emissions and the earmarking obligation would apply to the full revenue.

6 Implications of the aviation solidarity levy coalition mechanism for the EU budget

Under the proposed mechanism, the EU would not be able to retain all the aviation tax revenue for its budget. Recall that under the mechanism countries could at most retain the prescribed maximal retention rate, a design parameter that could reasonably set at e.g. 40%. In that case, the EU would have to earmark the remaining 60% of the tax revenue to global funds in proportion to how much funding they receive from all other countries. This would incentivize other countries to earmark money to their preferred global funds, given that this would mechanically crowd in some the EU's money.

Thus if the EU deems it a priority to increase its own resources, it could participate in the mechanism in this way with partial revenue retention. Alternatively, it could abstain from revenue retention but still indirectly increase its effectively available resources by replacing its current voluntary contributions to global funds by an equal funding stream from the aviation taxes. Currently, the EU gives 5 billion euros as voluntary budgetary contributions to global funds, including the Green Climate Fund, the Global Fund Against AIDS, TB and Malaria and the Global Alliance for Vaccines and Immunization. By instead funding these contributions via earmarking of aviation tax revenue[8], the EU could thus free up 5 billion euros for its budget.

In addition to freeing up resources for EU-internal spending, such a replacement of voluntary contributions by the earmarked aviation tax revenue would have at least two further beneficial effects. Firstly, it would make the funding stream resilient against potential pressure to discontinue it, particularly in times of rising nationalism. If such a solidarity levy coalition mechanism gets established and other countries join then the national gains from stopping earmarking to global funds altogether would greatly decline, given that in that case the other coalition members would automatically tax both all flights connecting them to the EU at the full rate instead of at half that rate, by the rules of the mechanism.

Secondly, there is a related potential benefit in terms of reduced support for nationalists, as we will now argue. Currently, nationalists commonly point towards voluntary contributions as evidence that governments are sacrificing their own population's welfare to benefit people in other countries. Factually, these claims are plausibly correct, at least in a narrow sense. Existing global funds like the Green Climate Fund, the Global Fund Against AIDS, TB and Malaria and the Global Alliance for Vaccines and Immunization mostly benefit the recipient countries, given that they either mostly crowd out domestic government spending or provide public benefits mostly accruing to the recipient countries (ref). Moreover, the design of these funds does not allow the donors to extract any concessions or return favors from the recipient countries, given that these funds' disbursement rules are fixed and universal. Precisely because of this, donating to global funds allows countries to gain some soft power and general goodwill from other countries, given that any donation signals that they have at least a little bit of altruistic motivation.

Despite the very small scale of these voluntary contributions, the frequent references to them by nationalists suggest that we cannot rule out that they might increase votes for nationalists. This poses a dilemma for governments: should they continue the voluntary contributions to the global funds or should they prioritize assuaging the fears of the part of the electorate responsive to the nationalist rhetoric that they do not get sufficiently prioritized?

In the context of the aviation solidarity levy coalition, this tension between (narrow) national interest and global welfare could decline. Initially, the EU could replace its current voluntary contributions to the global funds by earmarking of the revenue stream from the flight taxes. This would already decrease the force of the nationalist arguments, given that the opportunity cost of earmarking to such funds is small since only revenue retention of at most the retention rate (e.g. 40%) would be permitted. Moreover, as more and more countries join the aviation solidarity levy coalition mechanism global funds such as the Global Fund for AIDS, TB and Malaria would likely get funded via earmarking from the countries that most directly benefit from them, in particular if they manage to coordinate on such a decision in the context of the mechanism. The EU might then focus its earmarking on global funds that benefit it most, such as global funds for pandemic prevention and preparedness and global funds that reduce the global demand for oil (Edenhofer et al.).

[8]These global funds are likely to garner a population-weighted majority vote at the UN for eligibility for receiving earmarked funding via the aviation solidarity levy coalition.

Thus overall, the aviation levy coalition mechanism could better align national self-interest with global welfare and allow the EU to both contribute more to global public goods and free up money for EU-internal spending, while at the same time avoiding nationalist backlash that results from voluntary budgetary contributions to global funds.

Given the current 5 billion euros of voluntary contributions, the EU could thus free up this amount for its budget by substituting them via the earmarking from the flight taxes. The scope for such substitution could potentially be expanded if the EU were to manage to convince a sufficient set of other countries that specific EU programs and institutions are generating valuable global public goods such that they could become eligible (via the UN-based voting mechanism) for receiving earmarked tax revenue in the context of the aviation solidarity levy coalition mechanism.

A natural first candidate for an existing EU-funded program that could be nominated as a global fund for eligibility in the aviation levy coalition mechanism is the European & Developing Countries Clinical Trials Partnership (EDCTP). This is an EU-hosted institution to which the EU contributes EUR 0.16 billion per year. With a focus on global public good provision (research for vaccines, therapeutics, diagnostics, including for AMR resistance and pandemic preparedness) and a progressive transfer component via the strengthening of clinical research capacity in sub-Saharan Africa, it seems likely that a population-weighted majority vote would confer eligibility status to this global fund.

If and when such a fund becomes eligible via the UN-based vote, the EU could earmark aviation tax revenue to it, first replacing its current budgetary contributions and then expanding the funding stream beyond that. Given that parts of the benefits created by the EDCTP accrue to the EU (e.g. via reduced AMR resistance and pandemic preparedness innovation), the EU will plausibly have self-interested incentives to expand funding in the context of the aviation solidarity levy coalition mechanism, given that the opportunity cost of donating a euro would only be the retention rate parameter, e.g. 0.4 euros if the retention rate parameter is set at 0.4. This fact would make it in the interest of other countries to vote in favor of the EDCTP becoming eligible for funding via the aviation solidarity levy coalition mechanism. Again, overall the architecture helps to better align self-interest with global welfare, expanding both funding for global public goods and freeing up budgetary money for EU.

7 The (enlightened) self-interested case for the EU to initiate the aviation solidarity levy coalition mechanism

7.1 Avoiding international opposition and trade sanctions

The EU's experience in 2012 provides strong evidence on the political risks of unilateral extraterritorial aviation taxation with full revenue retention: Directive 2008/101/EC brought all flights arriving in or departing from EU airports into the ETS as of 1 January 2012. On 22 February 2012, twenty-three states adopted the Moscow Declaration, whose annex set out candidate countermeasures: proceedings under Article 84 of the Chicago Convention, prohibition of their own carriers from participating in the scheme, an assessment of the ETS against the EU's WTO commitments, review of air services agreements, suspension of negotiations on new routes, and additional levies on EU carriers. China's civil aviation authority had already, on 6 February 2012, forbidden Chinese carriers from participating without prior approval, and India followed. Airbus reported in March 2012 that thirty-five A330 and ten A380 orders — roughly $12 billion at list prices — were being withheld by China, and that it faced retaliation threats originating in twenty-six countries (EADS statements, March 2012). The Commission announced the "stop the clock" suspension on 12 November 2012, one day before the US Congress passed the EU ETS Prohibition Act, subsequently signed into law that month.[9] The measure had thus survived only eleven months.

The object of the opposition was not the carbon price as such but the combination of extraterritorial reach with full revenue retention. The incidence results in Section 2.3 explain why this combination is difficult to sustain. They show that the economic burden of an EU aviation carbon price falls substantially outside the EU — on travellers to and from the EU and on the hospitality sectors of the countries they visit — while

[9] The suspension was enacted as Decision No 377/2013/EU and has been extended by successive instruments; the reduced scope covering only intra-EEA flights remains in force today. See FIIA (2015) on the sequencing of the Commission's decision relative to the US legislation.

under full retention every euro of revenue stays inside the EU. From the standpoint of a third country, such a policy is a transfer, and opposing it is the materially rational response. Opposition in 2012 was therefore not an idiosyncratic overreaction to be discounted when forecasting the response to a future measure; it is what the model predicts.

To avoid such opposition, it seems essential to both make it in the material self-interest of other countries to accept the EU's extraterritorial taxation of flights, to create legitimacy for the policy by embedding it in a rules-based mechanism and to incentivise other countries to join such a mechanism. The proposed aviation solidarity levy coalition mechanism seems promising on all of these dimensions.

Concerning legitimacy, the proposed aviation solidarity levy coalition mechanism is designed to avoid discrimination objections, which played an important role in 2012. Under P^* the EU applies the full rate τ to its own domestic flights (P_0) while applying $\frac{\tau}{2}$ to international flights connecting to adopting countries; the elevated rate under P_2 to P_4 is conditioned on the policy of the partner country, not on the nationality or registration of the carrier. This is the design principle that the CBAM discussion suggests is decisive for avoiding legal difficulty, and it is the reason the proposal is constructed as a sequence of unilateral conditional policies rather than as membership conditionality.

7.2 Directing money to global funds particularly benefiting the EU

As a large player, the EU intenalises a substantial proportion of global public goods like pandemic prevention and climate change mitigation. Moreover, the EU can direct the aviation tax revenue to the specific global funds that particularly benefit the EU.

7.3 Inducing other countries to join the aviation solidarity levy coalition

The above results on the global incidence of carbon taxes on aviation reveal that there are strong unilateral incentives to introduce these taxes, even if countries are only allowed to retain 40% of the tax revenue. This would benefit the EU, both by crowding in earmarked contributions for global funds from other countries and via the reduction in aviation emissions in the rest of the world.

7.3.1 Crowding in earmarked contributions to global funds

By the definition of the allocation obligation, countries participating in the coalition mechanism will direct at least the fraction $1 - r$, where r denotes the retention rate parameter which could probably be set at $r = 0.4$ and still leave sufficiently strong incentives to motivate countries to participate. Moreover, countries that strongly prefer some global funds over others will find it optimal to forego their right to retain any tax revenue since by donating the entirety of their tax revenue they can also influence the allocation of the proportion $1 - r$ of the tax revenue collected by those who decide to retain the proportion r, by the rules of the mechanism.

7.3.2 Reducing aviation emissions in the rest of the world

This benefits the EU both via reduced climate damages and via reduced world market prices for oil.